

# SOLIDWORKS API (.NET VBA C#) — Reference

## (262/337): ISimulationMotorFeatureData- ISimulationSpringFeatureData

SOLIDWORKS API Documentation (converted from CHM)

# ISimulationMotorFeatureData (Type)

## ISimulationMotorFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationMotorFeatureData Interface

Allows access to the data that defines linear or rotary motors in SOLIDWORKS Motion studies.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface ISimulationMotorFeatureData
```

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData
```

C#

```
public interface ISimulationMotorFeatureData
```

C++/CLI

```
public interface class ISimulationMotorFeatureData
```

### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData](#).

### Example

[Create Linear Motor Feature \(VBA\)](#)  
[Fire Events for External Output Time Step Changes \(C#\)](#)  
[Fire Events for External Output Time Step Changes \(VB.NET\)](#)  
[Fire Events for External Output Time Step Changes \(VBA\)](#)

### Accessors

[IFeature::GetDefinition](#) and [IFeature::IGetDefinition](#)  
[IMotionStudy::CreateFeature](#)

### Access Diagram

[SimulationMotorFeatureData](#)

### See Also

[ISimulationMotorFeatureData Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulationGravityFeatureData Interface](#)  
[Simulation3DContactFeatureData Interface](#)  
[SimulationDamperFeatureData Interface](#)  
[SimulationForceFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

## ISimulationMotorFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationMotorFeatureData Interface

For a list of all members of this type, see [ISimulationMotorFeatureData members](#).

### Public Methods

| Name                                                                                                                   | Description                                                                        |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  <a href="#">ConstantSpeedMotor</a>   | Sets the constant speed for this motor feature.                                    |
|  <a href="#">DistanceMotor</a>        | Sets the distance and time for which to operate this motor feature.                |
|  <a href="#">GetInterpolatedValue</a> | Gets the interpolated value at the specified time for this motor feature.          |
|  <a href="#">InterpolatedMotor</a>    | Sets the drive type and interpolation scheme for this motor feature.               |
|  <a href="#">LoadSplineData</a>       | Loads the spline data from the specified file for this motor feature.              |
|  <a href="#">OscillatingMotor</a>     | Sets the displacement and frequency for oscillating motion for this motor feature. |

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## See Also

[ISimulationMotorFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulationGravityFeatureData Interface](#)  
[Simulation3DContactFeatureData Interface](#)  
[SimulationDamperFeatureData Interface](#)  
[SimulationForceFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

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## ISimulationMotorFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.sldworks Namespace](#) : ISimulationMotorFeatureData Interface

The following tables list the members exposed by [ISimulationMotorFeatureData](#).

## Public Properties

| Name                                                                                                                    | Description                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">DirectionReference</a>    | Gets or sets the direction in which the motor moves.                                                                                 |
|  <a href="#">DriveType</a>           | Sets the drive type of this motor feature.                                                                                           |
|  <a href="#">Expression</a>          | Gets or sets the motor motion expression for this motor feature.                                                                     |
|  <a href="#">ExternalState</a>       | Gets or sets whether your application can listen to motor-related motion study event.                                                |
|  <a href="#">InterpolationScheme</a> | Gets or set the interpolation scheme for this motor feature.                                                                         |
|  <a href="#">LoadReferences</a>      | Gets or sets the load references for this motor feature.                                                                             |
|  <a href="#">Location</a>            | Select a face, vertex, or edge on the assembly for the reference origin when setting motion relative to another part.                |
|  <a href="#">Magnitude</a>           | Get or set the magnitude for this motor feature.                                                                                     |
|  <a href="#">MotionType</a>          | Gets or sets the type of motion of this motor feature.                                                                               |
|  <a href="#">MotorType</a>           | Gets the type of motor for this motor feature.                                                                                       |
|  <a href="#">RelativeComponent</a>   | Gets or sets a part in the assembly to which to reference motion when setting motion relative to another part in this motor feature. |
|  <a href="#">ReverseDirection</a>    | Gets or sets whether or not to reverse the direction of the motor.                                                                   |
|  <a href="#">SplineData</a>          | Gets or sets the spline data points for this motor feature.                                                                          |
|  <a href="#">Velocity</a>            | Gets or sets the speed of the motor if no other force acts on it.                                                                    |

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## Public Methods

| Name                                                                                                                     | Description                                                                        |
|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  <a href="#">ConstantSpeedMotor</a>   | Sets the constant speed for this motor feature.                                    |
|  <a href="#">DistanceMotor</a>        | Sets the distance and time for which to operate this motor feature.                |
|  <a href="#">GetInterpolatedValue</a> | Gets the interpolated value at the specified time for this motor feature.          |
|  <a href="#">InterpolatedMotor</a>    | Sets the drive type and interpolation scheme for this motor feature.               |
|  <a href="#">LoadSplineData</a>       | Loads the spline data from the specified file for this motor feature.              |
|  <a href="#">OscillatingMotor</a>     | Sets the displacement and frequency for oscillating motion for this motor feature. |



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## See Also

[ISimulationMotorFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulationGravityFeatureData Interface](#)  
[Simulation3DContactFeatureData Interface](#)  
[SimulationDamperFeatureData Interface](#)  
[SimulationForceFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

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## ISimulationMotorFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : [ISimulationMotorFeatureData Interface](#)

For a list of all members of this type, see [ISimulationMotorFeatureData members](#).

## Public Properties

| Name                                                                                                                  | Description                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">DirectionReference</a>  | Gets or sets the direction in which the motor moves.                                                                                 |
|  <a href="#">DriveType</a>           | Sets the drive type of this motor feature.                                                                                           |
|  <a href="#">Expression</a>          | Gets or sets the motor motion expression for this motor feature.                                                                     |
|  <a href="#">ExternalState</a>       | Gets or sets whether your application can listen to motor-related motion study event.                                                |
|  <a href="#">InterpolationScheme</a> | Gets or set the interpolation scheme for this motor feature.                                                                         |
|  <a href="#">LoadReferences</a>      | Gets or sets the load references for this motor feature.                                                                             |
|  <a href="#">Location</a>           | Select a face, vertex, or edge on the assembly for the reference origin when setting motion relative to another part.                |
|  <a href="#">Magnitude</a>         | Get or set the magnitude for this motor feature.                                                                                     |
|  <a href="#">MotionType</a>        | Gets or sets the type of motion of this motor feature.                                                                               |
|  <a href="#">MotorType</a>         | Gets the type of motor for this motor feature.                                                                                       |
|  <a href="#">RelativeComponent</a> | Gets or sets a part in the assembly to which to reference motion when setting motion relative to another part in this motor feature. |
|  <a href="#">ReverseDirection</a>  | Gets or sets whether or not to reverse the direction of the motor.                                                                   |
|  <a href="#">SplineData</a>        | Gets or sets the spline data points for this motor feature.                                                                          |
|  <a href="#">Velocity</a>          | Gets or sets the speed of the motor if no other force acts on it.                                                                    |

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## See Also

[ISimulationMotorFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulationGravityFeatureData Interface](#)  
[Simulation3DContactFeatureData Interface](#)  
[SimulationDamperFeatureData Interface](#)  
[SimulationForceFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

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## ConstantSpeedMotor Method (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : ConstantSpeedMotor Method (ISimulationMotorFeatureData)

Sets the constant speed for this motor feature.

## .NET Syntax

Visual Basic (Declaration)

```
Sub ConstantSpeedMotor( _  
    ByVal Speed As System.Double _
```

)

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData
Dim Speed As System.Double

instance.ConstantSpeedMotor(Speed)
```

C#

```
void ConstantSpeedMotor(
    System.double Speed
)
```

C++/CLI

```
void ConstantSpeedMotor(
&    System.double Speed
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Speed*  
Constant speed in RPMs

### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::ConstantSpeedMotor](#).

### See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::DistanceMotor Method](#)  
[ISimulationMotorFeatureData::InterpolatedMotor Method](#)  
[ISimulationMotorFeatureData::OscillatingMotor Method](#)  
[ISimulationMotorFeatureData::Expression Property](#)

### Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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## DirectionReference Property (ISimulationMotorFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : DirectionReference Property (ISimulationMotorFeatureData)

Gets or sets the direction in which the motor moves.

### .NET Syntax

Visual Basic (Declaration)

```
Property DirectionReference As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData
Dim value As System.Object

instance.DirectionReference = value

value = instance.DirectionReference
```

C#

```
System.Object DirectionReference {get; set;}
```

C++/CLI

```
property System.Object^ DirectionReference {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Circular edge; a planar, cylindrical, or conical face; an axis; or a plane indicating the direction to move the motor

### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::DirectionReference](#).

### Example

[Create Linear Motor Feature \(VBA\)](#)

### See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::ReverseDirection Property](#)

### Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

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## DistanceMotor Method (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : DistanceMotor Method (ISimulationMotorFeatureData)

Sets the distance and time for which to operate this motor feature.

### .NET Syntax

Visual Basic (Declaration)

```
Sub DistanceMotor( _  
    ByVal Displacement As System.Double, _  
    ByVal StartTime As System.Double, _  
    ByVal Duration As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim Displacement As System.Double  
Dim StartTime As System.Double  
Dim Duration As System.Double
```

```
instance.DistanceMotor(Displacement, StartTime, Duration)
```

C#

```
void DistanceMotor(  
    System.double Displacement,  
    System.double StartTime,  
    System.double Duration  
)
```

C++/CLI

```

void DistanceMotor(
& System.double Displacement,
& System.double StartTime,
& System.double Duration
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Displacement*  
Displacement in degrees  
*StartTime*  
Start time in seconds  
*Duration*  
Duration in seconds

### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::DistanceMotor](#).

### See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::ConstantSpeedMotor Method](#)  
[ISimulationMotorFeatureData::InterpolatedMotor Method](#)  
[ISimulationMotorFeatureData::OscillatingMotor Method](#)  
[ISimulationMotorFeatureData::Expression Property](#)

### Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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## DriveType Property (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : DriveType Property (ISimulationMotorFeatureData)

Sets the drive type of this motor feature.

### .NET Syntax

#### Visual Basic (Declaration)

```
Property DriveType As System.Integer
```

#### Visual Basic (Usage)



```

Dim instance As ISimulationMotorFeatureData
Dim value As System.Integer

instance.DriveType = value

value = instance.DriveType

```

#### C#

```
System.int DriveType {get; set;}
```

#### C++/CLI

```

property System.int DriveType {
    System.int get();
    void set ( & System.int value);
}

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Drive type as defined in [swSimulationMotorDriveType\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::DriveType](#).

## See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)

## Availability

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## Expression Property (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : Expression Property (ISimulationMotorFeatureData)

Gets or sets the motor motion expression for this motor feature.

## .NET Syntax

Visual Basic (Declaration)

Property Expression As System.String

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim value As System.String
```

```
instance.Expression = value
```

```
value = instance.Expression
```

C#

```
System.string Expression {get; set;}
```

C++/CLI

```
property System.String^ Expression {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Motion motor expression in document units

## Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::Expression](#).

## See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::ConstantSpeedMotor Method](#)  
[ISimulationMotorFeatureData::DistanceMotor Method](#)  
[ISimulationMotorFeatureData::InterpolatedMotor Method](#)  
[ISimulationMotorFeatureData::OscillatingMotor Method](#)

## Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

## ExternalState Property (ISimulationMotorFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : ExternalState Property (ISimulationMotorFeatureData)

Gets or sets whether your application can listen to motor-related motion study event.

### .NET Syntax

Visual Basic (Declaration)

```
Property ExternalState As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim value As System.Boolean
```

```
instance.ExternalState = value
```

```
value = instance.ExternalState
```

C#

```
System.bool ExternalState {get; set;}
```

C++/CLI

```
property System.bool ExternalState {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True to listen to [IMotionStudy.MotorTimeStepChangeNotify](#) and [MotorOutputTimeStepChangeNotify](#), false to not

### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::ExternalState](#).

### Example

[Fire Events for External Output Time Step Changes \(C#\)](#)  
[Fire Events for External Output Time Step Changes \(VB.NET\)](#)  
[Fire Events for External Output Time Step Changes \(VBA\)](#)

### Remarks

When a motor's state is set to external, [IMotionStudy.MotorTimeStepChangeNotify](#) and [MotorOutputTimeStepChangeNotify](#) events are called with the motor name, time, and a pointer for the return value. These events are called at every major step from ADAMS. You can implement any motion law after catching these events. The return value for each event is a double that specifies the new motor speed.

### See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)

### Availability

SOLIDWORKS 2008 SP1, Revision Number 16.1

## GetInterpolatedValue Method (ISimulationMotorFeatureData)

[See Also](#)



Gets the interpolated value at the specified time for this motor feature.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetInterpolatedValue( _  
    ByVal Time As System.Double _  
) As System.Double
```

### Visual Basic (Usage)

```
Dim instance As ISimulationMotorFeatureData  
Dim Time As System.Double  
Dim value As System.Double  
  
value = instance.GetInterpolatedValue(Time)
```

### C#

```
System.double GetInterpolatedValue(  
    System.double Time  
)
```

### C++/CLI

```
System.double GetInterpolatedValue(  
& System.double Time  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Time*

Time for which you want the interpolated value

#### Return Value

Interpolated value at the specied time

## Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::GetInterpolatedValue](#).

## See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::InterpolatedMotor Method](#)  
[ISimulationMotorFeatureData::InterpolationScheme Property](#)

## Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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## InterpolatedMotor Method (ISimulationMotorFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : InterpolatedMotor Method (ISimulationMotorFeatureData)

Sets the drive type and interpolation scheme for this motor feature.

## .NET Syntax

### Visual Basic (Declaration)

```
Sub InterpolatedMotor( _  
    ByVal DriveType As System.Integer, _
```

```
ByVal InterpolationScheme As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim DriveType As System.Integer  
Dim InterpolationScheme As System.Integer  
  
instance.InterpolatedMotor(DriveType, InterpolationScheme)
```

C#

```
void InterpolatedMotor(  
    System.int DriveType,  
    System.int InterpolationScheme  
)
```

C++/CLI

```
void InterpolatedMotor(  
& System.int DriveType,  
& System.int InterpolationScheme  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*DriveType*  
Drive type as defined in [swSimulationMotorDriveType\\_e](#)  
*InterpolationScheme*  
Interpolation scheme

### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::InterpolatedMotor](#).

### Example

[Create Linear Motor Feature \(VBA\)](#)

### See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::GetInterpolatedValue Method](#)  
[ISimulationMotorFeatureData::InterpolationScheme Property](#)  
[ISimulationMotorFeatureData::ConstantSpeedMotor Method](#)  
[ISimulationMotorFeatureData::DistanceMotor Method](#)  
[ISimulationMotorFeatureData::OscillatingMotor Method](#)  
[ISimulationMotorFeatureData::Expression Property](#)

### Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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## InterpolationScheme Property (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : InterpolationScheme Property (ISimulationMotorFeatureData)

Gets or set the interpolation scheme for this motor feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property InterpolationScheme As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData
Dim value As System.Integer

instance.InterpolationScheme = value
value = instance.InterpolationScheme
```

C#

```
System.int InterpolationScheme {get; set;}
```

C++/CLI

```
property System.int InterpolationScheme {
    System.int get();
    void set ( &    System.int value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Interpolation scheme

### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::InterpolationScheme](#).

### See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::InterpolatedMotor Method](#)

### Availability

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## LoadReferences Property (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : LoadReferences Property (ISimulationMotorFeatureData)

Gets or sets the load references for this motor feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property LoadReferences As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData
Dim value As System.Object

instance.LoadReferences = value
value = instance.LoadReferences
```

C#

```
System.object LoadReferences {get; set;}
```

C++/CLI

```
property System.Object^ LoadReferences {
    System.Object^ get();
```

```
void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Load-bearing [faces](#) or [edges](#)

### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::LoadReferences](#).

### See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)

### Availability

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## LoadSplineData Method (ISimulationMotorFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : LoadSplineData Method (ISimulationMotorFeatureData)

Loads the spline data from the specified file for this motor feature.

### .NET Syntax

#### Visual Basic (Declaration)

```
Function LoadSplineData(  
    ByVal Value As System.String _  
) As System.Boolean
```

#### Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim Value As System.String  
Dim value As System.Boolean  
  
value = instance.LoadSplineData(Value)
```

#### C#

```
System.bool LoadSplineData(  
    System.string Value  
)
```

#### C++/CLI

```
System.bool LoadSplineData(  
    & System.String^ Value  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Value*  
File from which to load the spline data point values

#### Return Value

True if the operation succeeds, false if it fails

### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::LoadSplineData](#).

## Example

[Create Linear Motor Feature \(VBA\)](#)

## Remarks

The file should contain one data point per line. The data point consists of two values:

- Time
- Value at that time

Use commas or spaces as separators between the values.

## See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::SplineData Property](#)

## Availability

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## Location Property (ISimulationMotorFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : Location Property (ISimulationMotorFeatureData)

Select a face, vertex, or edge on the assembly for the reference origin when setting motion relative to another part.

## .NET Syntax

Visual Basic (Declaration)

Property Location As System.Object

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData
Dim value As System.Object

instance.Location = value

value = instance.Location
```

C#

```
System.Object Location {get; set;}
```

C++/CLI

```
property System.Object^ Location {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

[Face](#), [vertex](#), or [edge](#)

## Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::Location](#).

## Example

[Create Linear Motor Feature \(VBA\)](#)

## See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::RelativeComponent Property](#)  
[ISimulationMotorFeatureData::LoadReferences Property](#)

## Availability

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## Magnitude Property (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : Magnitude Property (ISimulationMotorFeatureData)

Get or set the magnitude for this motor feature.

## .NET Syntax

Visual Basic (Declaration)

Property Magnitude As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim value As System.Double
```

```
instance.Magnitude = value
```

```
value = instance.Magnitude
```

C#

```
System.double Magnitude {get; set;}
```

C++/CLI

```
property System.double Magnitude {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Magnitude

## Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::Magnitude](#).

## See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)

## Availability

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## MotionType Property (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : MotionType Property (ISimulationMotorFeatureData)

Gets or sets the type of motion of this motor feature.

## Visual Basic Syntax

### Visual Basic (Declaration)

```
Property MotionType As System.Integer
```

### Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim value As System.Integer  
  
instance.MotionType = value  
value = instance.MotionType
```

### C#

```
System.int MotionType {get; set;}
```

### C++/CLI

```
property System.int MotionType {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Type of motion as defined in [swSimulationMotorMotionType\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::MotionType](#).

## See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)

## Availability

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### MotorType Property (ISimulationMotorFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : MotorType Property (ISimulationMotorFeatureData)

Gets the type of motor for this motor feature.

## Visual Basic Syntax

### Visual Basic (Declaration)

```
ReadOnly Property MotorType As System.Integer
```

### Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim value As System.Integer  
  
value = instance.MotorType
```

### C#

```
System.int MotorType {get;}
```

C++/CLI

```
property System::int MotorType {  
    System::int get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Type of motor as defined in [swSimulationMotorType\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::MotorType](#).

### Example

[Create Linear Motor Feature \(VBA\)](#)

### See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)

### Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

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## OscillatingMotor Method (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : OscillatingMotor Method (ISimulationMotorFeatureData)

Sets the displacement and frequency for oscillating motion for this motor feature.

### .NET Syntax

Visual Basic (Declaration)

```
Sub OscillatingMotor( _  
    ByVal Displacement As System.Double, _  
    ByVal Frequency As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim Displacement As System.Double  
Dim Frequency As System.Double  
  
instance.OscillatingMotor(Displacement, Frequency)
```

C#

```
void OscillatingMotor(  
    System.double Displacement,  
    System.double Frequency  
)
```

C++/CLI

```
void OscillatingMotor(  
& System::double Displacement,  
& System::double Frequency  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Displacement*  
Displacement in degrees  
*Frequency*  
Frequency in Hz

## Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::OscillatingMotor](#).

## See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::ConstantSpeedMotor Method](#)  
[ISimulationMotorFeatureData::DistanceMotor Method](#)  
[ISimulationMotorFeatureData::InterpolatedMotor Method](#)  
[ISimulationMotorFeatureData::Expression Property](#)

## Availability

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## RelativeComponent Property (ISimulationMotorFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : RelativeComponent Property (ISimulationMotorFeatureData)

Gets or sets a part in the assembly to which to reference motion when setting motion relative to another part in this motor feature.

## .NET Syntax

Visual Basic (Declaration)

Property RelativeComponent As Component2

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim value As Component2
```

```
instance.RelativeComponent = value
```

```
value = instance.RelativeComponent
```

C#

```
Component2 RelativeComponent {get; set;}
```

C++/CLI

```
property Component2^ RelativeComponent {  
    Component2^ get();  
    void set ( & Component2^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Relative [component](#)

## Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::RelativeComponent](#).

## Example

[Create Linear Motor Feature \(VBA\)](#)

## See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::Location Property](#)  
[ISimulationMotorFeatureData::LoadReferences Property](#)

## Availability

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## ReverseDirection Property (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : ReverseDirection Property (ISimulationMotorFeatureData)

Gets or sets whether or not to reverse the direction of the motor.

## .NET Syntax

Visual Basic (Declaration)

```
Property ReverseDirection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationMotorFeatureData  
Dim value As System.Boolean
```

```
instance.ReverseDirection = value
```

```
value = instance.ReverseDirection
```

C#

```
System.bool ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool ReverseDirection {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to reverse the direction of the motor, false to not

## Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::ReverseDirection](#).

## See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::DirectionReference Property](#)

## Availability

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## SplineData Property (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : SplineData Property (ISimulationMotorFeatureData)

Gets or sets the spline data points for this motor feature.

 **.NET Syntax**

Visual Basic (Declaration)

Property SplineData As System.Object

Visual Basic (Usage)

Dim instance As [ISimulationMotorFeatureData](#)  
Dim value As System.Object  
  
instance.SplineData = value  
  
value = instance.SplineData

C#

System.object SplineData {get; set;}

C++/CLI

property System.Object^ SplineData {  
 System.Object^ get();  
 void set ( & System.Object^ value);  
}

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of spline data points

 **Visual Basic for Applications (VBA) Syntax**

See [SimulationMotorFeatureData::SplineData](#).

 **See Also**

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)  
[ISimulationMotorFeatureData::LoadSplineData Method](#)

 **Availability**

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Velocity Property (ISimulationMotorFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationMotorFeatureData Interface](#) : Velocity Property (ISimulationMotorFeatureData)

Gets or sets the speed of the motor if no other force acts on it.

 **.NET Syntax**

Visual Basic (Declaration)

Property Velocity As System.Double

Visual Basic (Usage)

```
Dim instance As ISimulationMotorFeatureData  
Dim value As System.Double  
  
instance.Velocity = value  
  
value = instance.Velocity
```

C#

```
System.double Velocity {get; set;}
```

C++/CLI

```
property System.double Velocity {  
    System.double get();  
    void set ( &    System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Speed of the motor

#### Visual Basic for Applications (VBA) Syntax

See [SimulationMotorFeatureData::Velocity](#).

#### See Also

[ISimulationMotorFeatureData Interface](#)  
[ISimulationMotorFeatureData Members](#)

#### Availability

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# ISimulationSpringFeatureData (Type)

## ISimulationSpringFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationSpringFeatureData Interface

Allows access to the data that defines a simulation spring feature in SOLIDWORKS Motion studies.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface ISimulationSpringFeatureData
```

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData
```

C#

```
public interface ISimulationSpringFeatureData
```

C++/CLI

```
public interface class ISimulationSpringFeatureData
```

### Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData](#).

### Example

[Add Spring to Motion Study \(C#\)](#)  
[Add Spring to Motion Study \(VB.NET\)](#)  
[Add Spring to Motion Study \(VBA\)](#)

### Remarks

Motion due to motors supersedes motion due to springs. If you have a motor moving a component to the left and a spring pulling a component to the right, the component moves to the left, and the power consumption of the motor increases.

### Accessors

[IFeature::GetDefinition](#) and [IFeature::IGetDefinition](#)  
[IMotionStudy::CreateFeature](#)

### Access Diagram

[SimulationSpringFeatureData](#)

### See Also

[ISimulationSpringFeatureData Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulation3DContactFeatureData Interface](#)  
[ISimulationDamperFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)

## ISimulationSpringFeatureData Interface Methods

[See Also](#)



For a list of all members of this type, see [ISimulationSpringFeatureData members](#).

Public Methods

| Name                                                                                                           | Description                                             |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
|  <a href="#">GetEndPoints</a> | Gets the end points for this simulation spring feature. |
|  <a href="#">SetEndPoints</a> | Sets the end points for this simulation spring feature. |

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See Also

[ISimulationSpringFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulation3DContactFeatureData Interface](#)  
[ISimulationDamperFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)

ISimulationSpringFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)

The following tables list the members exposed by [ISimulationSpringFeatureData](#).

Public Properties

| Name                                                                                                                                | Description                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">BaseComponent</a>                     | Gets or sets the base component for this simulation spring feature.                                                                             |
|  <a href="#">CoilDiameter</a>                     | Gets or sets the diameter of the coil for this simulation spring feature.                                                                       |
|  <a href="#">DampingConstant</a>                 | Gets or sets the damping constant for this simulation spring feature.                                                                           |
|  <a href="#">ExponentOfDamperForceExpression</a> | Gets or sets the exponent of the damper force expression for this simulation spring feature.                                                    |
|  <a href="#">ExponentOfSpringForceExpression</a> | Gets or sets the exponent of the spring force expression for this simulation spring feature.                                                    |
|  <a href="#">FreeAngle</a>                       | Gets or sets the free angle for the functional expression for this simulation spring feature.                                                   |
|  <a href="#">FreeLength</a>                      | Gets or sets the free length (i.e., length to stretch or compress the spring) for the functional expression for this simulation spring feature. |
|  <a href="#">HasDamper</a>                       | Gets or sets whether this simulation spring feature has damper.                                                                                 |
|  <a href="#">LoadReferences</a>                  | Gets or sets the load references for this simulation spring feature.                                                                            |
|  <a href="#">NumberOfCoils</a>                   | Gets or sets the number of coils in this simulation spring feature.                                                                             |
|  <a href="#">ReverseDirection</a>                | Gets or sets whether to reverse the direction of this simulation spring feature.                                                                |
|  <a href="#">SpringConstant</a>                  | Gets or sets the strength of this simulation spring feature.                                                                                    |
|  <a href="#">Type</a>                            | Gets or sets the type of simulation spring feature.                                                                                             |
|  <a href="#">UpdateToModelChanges</a>            | Gets or sets whether to have the <a href="#">free length</a> dynamically update model changes while the PropertyManager page is open.           |
|  <a href="#">WireDiameter</a>                    | Gets or sets the diameter of the wire for this simulation spring feature.                                                                       |

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Public Methods

| Name                                                                                                             | Description                                             |
|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
|  <a href="#">GetEndPoints</a> | Gets the end points for this simulation spring feature. |
|  <a href="#">SetEndPoints</a> | Sets the end points for this simulation spring feature. |

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See Also

[ISimulationSpringFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulation3DContactFeatureData Interface](#)  
[ISimulationDamperFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)

ISimulationSpringFeatureData Interface Properties

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : ISimulationSpringFeatureData Interface

For a list of all members of this type, see [ISimulationSpringFeatureData members](#).

 **Public Properties**

| Name                                                                                                                              | Description                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">BaseComponent</a>                   | Gets or sets the base component for this simulation spring feature.                                                                             |
|  <a href="#">CoilDiameter</a>                    | Gets or sets the diameter of the coil for this simulation spring feature.                                                                       |
|  <a href="#">DampingConstant</a>                 | Gets or sets the damping constant for this simulation spring feature.                                                                           |
|  <a href="#">ExponentOfDamperForceExpression</a> | Gets or sets the exponent of the damper force expression for this simulation spring feature.                                                    |
|  <a href="#">ExponentOfSpringForceExpression</a> | Gets or sets the exponent of the spring force expression for this simulation spring feature.                                                    |
|  <a href="#">FreeAngle</a>                       | Gets or sets the free angle for the functional expression for this simulation spring feature.                                                   |
|  <a href="#">FreeLength</a>                      | Gets or sets the free length (i.e., length to stretch or compress the spring) for the functional expression for this simulation spring feature. |
|  <a href="#">HasDamper</a>                       | Gets or sets whether this simulation spring feature has damper.                                                                                 |
|  <a href="#">LoadReferences</a>                  | Gets or sets the load references for this simulation spring feature.                                                                            |
|  <a href="#">NumberOfCoils</a>                   | Gets or sets the number of coils in this simulation spring feature.                                                                             |
|  <a href="#">ReverseDirection</a>                | Gets or sets whether to reverse the direction of this simulation spring feature.                                                                |
|  <a href="#">SpringConstant</a>                | Gets or sets the strength of this simulation spring feature.                                                                                    |
|  <a href="#">Type</a>                          | Gets or sets the type of simulation spring feature.                                                                                             |
|  <a href="#">UpdateToModelChanges</a>          | Gets or sets whether to have the <a href="#">free length</a> dynamically update model changes while the PropertyManager page is open.           |
|  <a href="#">WireDiameter</a>                  | Gets or sets the diameter of the wire for this simulation spring feature.                                                                       |

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 **See Also**

[ISimulationSpringFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulation3DContactFeatureData Interface](#)  
[ISimulationDamperFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)

BaseComponent Property (ISimulationSpringFeatureData)

[See Also](#)

  [SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : BaseComponent Property (ISimulationSpringFeatureData)

Gets or sets the base component for this simulation spring feature.

 **Visual Basic Syntax**

Visual Basic (Declaration)

Property BaseComponent As Component2

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As Component2
```

```
instance.BaseComponent = value  
value = instance.BaseComponent
```

C#

```
Component2 BaseComponent {get; set;}
```

C++/CLI

```
property Component2^ BaseComponent {  
    Component2^ get();  
    void set ( & Component2^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

[Base component](#)

### Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::BaseComponent](#).

### See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)

### Availability

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## CoilDiameter Property (ISimulationSpringFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : CoilDiameter Property (ISimulationSpringFeatureData)

Gets or sets the diameter of the coil for this simulation spring feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property CoilDiameter As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As System.Double
```

```
instance.CoilDiameter = value
```

```
value = instance.CoilDiameter
```

C#

```
System.double CoilDiameter {get; set;}
```

C++/CLI

```
property System.double CoilDiameter {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Property Value

Coil diameter

## Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::CoilDiameter](#).

## Example

[Add Spring to Motion Study \(C#\)](#)  
[Add Spring to Motion Study \(VB.NET\)](#)  
[Add Spring to Motion Study \(VBA\)](#)

## See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::NumberOfCoils Property](#)  
[ISimulationSpringFeatureData::WireDiameter Property](#)

## Availability

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## DampingConstant Property (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : DampingConstant Property (ISimulationSpringFeatureData)

Gets or sets the damping constant for this simulation spring feature.

## .NET Syntax

Visual Basic (Declaration)

Property DampingConstant As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As System.Double  
  
instance.DampingConstant = value  
value = instance.DampingConstant
```

C#

```
System.double DampingConstant {get; set;}
```

C++/CLI

```
property System.double DampingConstant {  
    System.double get();  
    void set ( &    System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Property Value

Damping constant

## Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::DampingConstant](#).

## See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::HasDamper Property](#)  
[ISimulationSpringFeatureData::ExponentOfDamperForceExpression Property](#)

## Availability

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## ExponentOfDamperForceExpression Property (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : ExponentOfDamperForceExpression Property (ISimulationSpringFeatureData)

Gets or sets the exponent of the damper force expression for this simulation spring feature.

## .NET Syntax

### Visual Basic (Declaration)

Property ExponentOfDamperForceExpression As System.Integer

### Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As System.Integer  
  
instance.ExponentOfDamperForceExpression = value  
value = instance.ExponentOfDamperForceExpression
```

### C#

```
System.int ExponentOfDamperForceExpression {get; set;}
```

### C++/CLI

```
property System.int ExponentOfDamperForceExpression {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Exponent of the damper force expression

## Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::ExponentOfDamperForceExpression](#).

## See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::DampingConstant Property](#)  
[ISimulationSpringFeatureData::HasDamper Property](#)  
[ISimulationSpringFeatureData::FreeAngle Property](#)  
[ISimulationSpringFeatureData::FreeLength Property](#)  
[ISimulationSpringFeatureData::SpringConstant Property](#)

## Availability

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## ExponentOfSpringForceExpression Property (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : ExponentOfSpringForceExpression Property (ISimulationSpringFeatureData)

Gets or sets the exponent of the spring force expression for this simulation spring feature.

## .NET Syntax

### Visual Basic (Declaration)

Property ExponentOfSpringForceExpression As System.Integer

### Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As System.Integer  
  
instance.ExponentOfSpringForceExpression = value  
value = instance.ExponentOfSpringForceExpression
```

### C#

```
System.int ExponentOfSpringForceExpression {get; set;}
```

### C++/CLI

```
property System.int ExponentOfSpringForceExpression {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Exponent of the spring force expression

## Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::ExponentOfSpringForceExpression](#).

## See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::SpringConstant Property](#)  
[ISimulationSpringFeatureData::FreeAngle Property](#)  
[ISimulationSpringFeatureData::FreeLength Property](#)

## Availability

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## FreeAngle Property (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : FreeAngle Property (ISimulationSpringFeatureData)

Gets or sets the free angle for the functional expression for this simulation spring feature.

## .NET Syntax

### Visual Basic (Declaration)

Property FreeAngle As System.Double

### Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData
Dim value As System.Double

instance.FreeAngle = value

value = instance.FreeAngle
```

C#

```
System.double FreeAngle {get; set;}
```

C++/CLI

```
property System.double FreeAngle {
    System.double get();
    void set ( &    System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Free angle

### Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::FreeAngle](#).

### See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::ExponentOfDamperForceExpression Property](#)  
[ISimulationSpringFeatureData::ExponentOfSpringForceExpression Property](#)  
[ISimulationSpringFeatureData::FreeLength Property](#)  
[ISimulationSpringFeatureData::SpringConstant Property](#)

### Availability

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## FreeLength Property (ISimulationSpringFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : FreeLength Property (ISimulationSpringFeatureData)

Gets or sets the free length (i.e., length to stretch or compress the spring) for the functional expression for this simulation spring feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property FreeLength As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData
Dim value As System.Double

instance.FreeLength = value

value = instance.FreeLength
```

C#

```
System.double FreeLength {get; set;}
```

C++/CLI

```
property System.double FreeLength {
    System.double get();
    void set ( & System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Free length

### Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::FreeLength](#).

### Example

[Add Spring to Motion Study \(C#\)](#)  
[Add Spring to Motion Study \(VB.NET\)](#)  
[Add Spring to Motion Study \(VBA\)](#)

### Remarks

The initial distance is the distance between the parts as currently displayed in the graphics area. Call [ISimulationSpringFeatureData::UpdateToModelChanges](#) to have the free length dynamically update to model changes while the PropertyManager page is open.

The spring does not exert any force when its length is equal to its free length.

### See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::ExponentOfDamperForceExpression Property](#)  
[ISimulationSpringFeatureData::ExponentOfSpringForceExpression Property](#)  
[ISimulationSpringFeatureData::FreeAngle Property](#)  
[ISimulationSpringFeatureData::SpringConstant Property](#)

### Availability

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## GetEndPoints Method (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : GetEndPoints Method (ISimulationSpringFeatureData)

Gets the end points for this simulation spring feature.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetEndPoints( _
    ByRef PDirDisp1 As System.Object, _
    ByRef PDirDisp2 As System.Object, _
    ByRef Type1 As System.Integer, _
    ByRef Type2 As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData
Dim PDirDisp1 As System.Object
Dim PDirDisp2 As System.Object
Dim Type1 As System.Integer
Dim Type2 As System.Integer
Dim value As System.Boolean
```

```
value = instance.GetEndPoints(PDirDisp1, PDirDisp2, Type1, Type2)
```

C#

```
System.bool GetEndPoints(  
    out System.object PDirDisp1,  
    out System.object PDirDisp2,  
    out System.int Type1,  
    out System.int Type2  
)
```

C++/CLI

```
System.bool GetEndPoints(  
    & [Out] System.Object^ PDirDisp1,  
    & [Out] System.Object^ PDirDisp2,  
    & [Out] System.int Type1,  
    & [Out] System.int Type2  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*PDirDisp1*

Feature

*PDirDisp2*

Feature

*Type1*

Type of end point as defined in [swSelectType\\_e](#)

*Type2*

Type of end point as defined in [swSelectType\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::GetEndPoints](#).

### See Also

[ISimulationSpringFeatureData Interface](#)

[ISimulationSpringFeatureData Members](#)

[ISimulationSpringFeatureData::SetEndPoints Method](#)

### Availability

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## HasDamper Property (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : HasDamper Property (ISimulationSpringFeatureData)

Gets or sets whether this simulation spring feature has damper.

### .NET Syntax

Visual Basic (Declaration)

```
Property HasDamper As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData
```

```
Dim value As System.Boolean
```

```
instance.HasDamper = value
```

```
value = instance.HasDamper
```

C#

```
System.bool HasDamper {get; set;}
```

C++/CLI

```
property System.bool HasDamper {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True if this simulation spring feature has damper, false if not

### Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::HasDamper](#).

### See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::DampingConstant Property](#)  
[ISimulationSpringFeatureData::ExponentOfDamperForceExpression Property](#)

### Availability

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## LoadReferences Property (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : LoadReferences Property (ISimulationSpringFeatureData)

Gets or sets the load references for this simulation spring feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property LoadReferences As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As System.Object
```

```
instance.LoadReferences = value
```

```
value = instance.LoadReferences
```

C#

```
System.object LoadReferences {get; set;}
```

C++/CLI

```
property System.Object^ LoadReferences {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Load-bearing [faces](#)

## Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::LoadReferences](#).

## See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)

## Availability

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## NumberOfCoils Property (ISimulationSpringFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : NumberOfCoils Property (ISimulationSpringFeatureData)

Gets or sets the number of coils in this simulation spring feature.

## .NET Syntax

Visual Basic (Declaration)

Property NumberOfCoils As System.Integer

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As System.Integer
```

```
instance.NumberOfCoils = value
```

```
value = instance.NumberOfCoils
```

C#

```
System.int NumberOfCoils {get; set;}
```

C++/CLI

```
property System.int NumberOfCoils {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Number of coils

## Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::NumberOfCoils](#).

## Example

[Add Spring to Motion Study \(C#\)](#)  
[Add Spring to Motion Study \(VB.NET\)](#)  
[Add Spring to Motion Study \(VBA\)](#)

## See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::CoilDiameter Property](#)  
[ISimulationSpringFeatureData::WireDiameter Property](#)

## Availability

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## ReverseDirection Property (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : ReverseDirection Property (ISimulationSpringFeatureData)

Gets or sets whether to reverse the direction of this simulation spring feature.

## .NET Syntax

Visual Basic (Declaration)

Property ReverseDirection As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData
Dim value As System.Boolean
```

```
instance.ReverseDirection = value
```

```
value = instance.ReverseDirection
```

C#

```
System.bool ReverseDirection {get; set;}
```

C++/CLI

```
property System.bool ReverseDirection {
    System.bool get();
    void set ( & System.bool value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to reverse the direction of this simulation spring feature, false to not

## Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::ReverseDirection](#).

## See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)

## Availability

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## SetEndPoints Method (ISimulationSpringFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : SetEndPoints Method (ISimulationSpringFeatureData)

Sets the end points for this simulation spring feature.

## .NET Syntax

Visual Basic (Declaration)

```
Function SetEndPoints( _  
    ByVal PDirDisp1 As System.Object, _  
    ByVal PDirDisp2 As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim PDirDisp1 As System.Object  
Dim PDirDisp2 As System.Object  
Dim value As System.Boolean  
  
value = instance.SetEndPoints(PDirDisp1, PDirDisp2)
```

C#

```
System.bool SetEndPoints(  
    System.object PDirDisp1,  
    System.object PDirDisp2  
)
```

C++/CLI

```
System.bool SetEndPoints(  
& System.Object^ PDirDisp1,  
& System.Object^ PDirDisp2  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*PDirDisp1*  
Feature  
*PDirDisp2*  
Feature

### Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::SetEndPoints](#).

### Example

[Add Spring to Motion Study \(C#\)](#)  
[Add Spring to Motion Study \(VB.NET\)](#)  
[Add Spring to Motion Study \(VBA\)](#)

### See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::GetEndPoints Method](#)

### Availability

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## SpringConstant Property (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : SpringConstant Property (ISimulationSpringFeatureData)

Gets or sets the strength of this simulation spring feature.

### .NET Syntax

Visual Basic (Declaration)

## Property SpringConstant As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As System.Double  
  
instance.SpringConstant = value  
value = instance.SpringConstant
```

C#

```
System.double SpringConstant {get; set;}
```

C++/CLI

```
property System.double SpringConstant {  
    System.double get();  
    void set ( &    System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Strength

## Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::SpringConstant](#).

## Remarks

Springs apply a force to a component. A spring with a higher spring constant moves a component faster than a spring with a lower spring constant. A component with a smaller mass moves faster than a component with a larger mass when acted upon by an equal strength spring.

## See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::ExponentOfDamperForceExpression Property](#)  
[ISimulationSpringFeatureData::ExponentOfSpringForceExpression Property](#)  
[ISimulationSpringFeatureData::FreeAngle Property](#)  
[ISimulationSpringFeatureData::FreeLength Property](#)

## Availability

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## Type Property (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : Type Property (ISimulationSpringFeatureData)

Gets or sets the type of simulation spring feature.

## .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Type As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As System.Integer
```

```
value = instance.Type
```

C#

```
System.int Type {get;}
```

C++/CLI

```
property System.int Type {  
    System.int get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Spring type as defined by [swSpringType\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::Type](#).

### Example

See [ISimulationSpringFeatureData](#) examples.

### See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)

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## UpdateToModelChanges Property (ISimulationSpringFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : UpdateToModelChanges Property (ISimulationSpringFeatureData)

Gets or sets whether to have the [free length](#) dynamically update model changes while the PropertyManager page is open.

### .NET Syntax

Visual Basic (Declaration)

```
Property UpdateToModelChanges As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As System.Boolean
```

```
instance.UpdateToModelChanges = value
```

```
value = instance.UpdateToModelChanges
```

C#

```
System.bool UpdateToModelChanges {get; set;}
```

C++/CLI

```
property System.bool UpdateToModelChanges {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Ture to have the free length dynamically update model changes while the PropertyManager page is open, false to not

## Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::UpdateToModelChanges](#).

## See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)

## Availability

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## WireDiameter Property (ISimulationSpringFeatureData)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationSpringFeatureData Interface](#) : WireDiameter Property (ISimulationSpringFeatureData)

Gets or sets the diameter of the wire for this simulation spring feature.

## .NET Syntax

Visual Basic (Declaration)

Property WireDiameter As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulationSpringFeatureData  
Dim value As System.Double
```

```
instance.WireDiameter = value
```

```
value = instance.WireDiameter
```

C#

```
System.double WireDiameter {get; set;}
```

C++/CLI

```
property System.double WireDiameter {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Wire diameter

## Visual Basic for Applications (VBA) Syntax

See [SimulationSpringFeatureData::WireDiameter](#).

## Example

[Add Spring to Motion Study \(C#\)](#)  
[Add Spring to Motion Study \(VB.NET\)](#)  
[Add Spring to Motion Study \(VBA\)](#)

## See Also

[ISimulationSpringFeatureData Interface](#)  
[ISimulationSpringFeatureData Members](#)  
[ISimulationSpringFeatureData::CoilDiameter Property](#)  
[ISimulationSpringFeatureData::NumberOfCoils Property](#)

